4 The gravitational field strength at the surface of the Earth varies slightly from place to place.

At the South Pole it is 9.86 N / kg and in London it is 9.81 N / kg.

An explorer collects a rock at the South Pole and moves it to London.

How does moving the rock from the South Pole to London affect the mass and the weight of the rock?

	effect on mass	**effect on weight**
A	none	decreases
B	none	none
C	decreases	decreases
D	decreases	none